Jean Pauphilet

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Academic Appointments

London Business School — London, United Kingdom
Associate Professor (with tenure) — Apr 2026–
Assistant Professor — Aug 2020–Apr 2026

Research Interests

Methods	Discrete optimization, machine learning, robust optimization, statistics
Applications	Healthcare, hospital operations, sustainable operations, social impact

Education

Massachusetts Institute of Technology — Cambridge, MA
PhD. in Operations Research, Analytics track (advisor: Dimitris Bertsimas) — Sep 2016–May 2020
2020 INFORMS Dantzig Dissertation Award, Finalist

École Polytechnique – Corps des Mines — Paris, France
Diplôme d'ingénieur (MSc. equivalent) — Sep 2012–Aug 2015

Work Under Review

Algorithms for discrete and robust optimization

- *PaNGEA: Parallel Node Generation and Exploration Algorithm*
  with Y. Wu
- *Compact Lifted Relaxations for Low-Rank Optimization*
  with R. Cory-Wright
- *Improved Approximation Algorithms for Orthogonally Constrained Problems Using Semidefinite Optimization*
  with R. Cory-Wright
  - Preliminary version accepted to IPCO 2026
- *Generating a Diverse Set of Near-Optimal Solutions: Why and How*
  with D. den Hertog, B. Song, J. Starreveld
- *The Surprising Performance of Random Partial Benders Decomposition*
  with Y. Wu
  - (Y. Wu) 2025 INFORMS Computing Society Student Prize, Honorable Mention
- *Disjunctive Branch-And-Bound for Certifiably Optimal Low-Rank Matrix Completion*
  with D. Bertsimas, R. Cory-Wright, and S. Lo

Optimization for machine learning and statistics

- *msPCA: An R Package for Sparse PCA with Multiple Components*
  with R. Cory-Wright

Analytics for healthcare operations

- *Occupancy Prediction with Patient Data: Evaluating Time-Series, Patient-Level Aggregation, and Deep Set Models*
  with S-H. Kim and W. Overman

Analytics for sustainable operations

- *Operationalizing Experimental Design: Data Collection for Remote Ocean Monitoring*
  with B. Song

Publications

Algorithms for discrete and robust optimization

25. *A Stochastic Benders Decomposition Scheme for Large-Scale Stochastic Network Design*
    with D. Bertsimas, R. Cory-Wright, and P. Petridis
    INFORMS Journal on Computing, 2024

24. *Minkowski Centers via Robust Optimization: Computation and Applications*
    with D. den Hertog and M.Y. Soali
    Operations Research, 2023

23. *A New Perspective on Low-Rank Optimization*
    with D. Bertsimas and R. Cory-Wright
    Mathematical Programming, 2023

22. *Robust Convex Optimization: A New Perspective That Unifies And Extends*
    with D. Bertsimas, D. den Hertog, and J. Zhen
    Mathematical Programming, 2022

21. *Mixed-Projection Conic Optimization: A New Paradigm for Modeling Rank Constraints*
    with D. Bertsimas and R. Cory-Wright
    Operations Research, 2021
    - 2020 INFORMS George E. Nicholson Student Paper Competition, Winner

20. *Probabilistic Guarantees in Robust Optimization*
    with D. Bertsimas and D. den Hertog
    SIAM Journal on Optimization, 2021

19. *A Unified Approach to Mixed-Integer Optimization: Nonlinear Formulations and Scalable Algorithms*
    with D. Bertsimas and R. Cory-Wright
    SIAM Journal on Optimization, 2021
    - 2020 Mixed Integer Programming Workshop Student Poster Competition, Finalist
    - 2019 INFORMS Computing Society Student Paper Award, Winner

Optimization for machine learning and statistics

18. *Sparse PCA with Multiple Components*
    with R. Cory-Wright
    Operations Research, 2026
    - 2024 INFORMS Data Mining and Decision Analytics Workshop Theoretical Paper Competition, Winner

17. *Adaptive Optimization for Prediction with Missing Data*
with D. Bertsimas and A. Delarue
Machine Learning, 2025

16. *Simple Imputation Rules for Prediction with Missing Data: Theoretical Guarantees vs. Empirical Performance*
with D. Bertsimas and A. Delarue
Transactions on Machine Learning Research, 2024
    - 2021 INFORMS Data Mining Section Best Paper, Finalist

15. *Robust and Heterogeneous Odds Ratio: Estimating Price Sensitivity for Unbought Items*
Manufacturing & Service Operations Management, 2022

14. *Solving Large-Scale Sparse PCA to Certifiable (Near) Optimality*
with D. Bertsimas and R. Cory-Wright
Journal of Machine Learning Research, 2022

13. *Sparse Classification: A Scalable Discrete Optimization Perspective*
with D. Bertsimas and B. Van Parys
Machine Learning, 2021

12. *Sparse Regression: Scalable Algorithms and Empirical Performance*
with D. Bertsimas and B. Van Parys
Statistical Science, 2020

11. *Certifiably Optimal Sparse Inverse Covariance Estimation*
with D. Bertsimas and J. Lamperski
Mathematical Programming, 2020

Analytics for healthcare operations

10. *Patient Outcome Predictions Improve Operations at Hartford HealthCare*
with D. Bertsimas, L. Na, K. Villalobos Carballo, and Hartford HealthCare
INFORMS Journal on Applied Analytics, 2025
    - (L. Na and K. Villalobos Carballo) 2023 INFORMS Doing Good with Good OR Student Competition, Finalist

9. *Optimization Automates Nurse Scheduling at Hartford Hospital Emergency Department*
with D. Bertsimas, L. Na, and Hartford HealthCare
INFORMS Journal on Applied Analytics, 2024

8. *The Best Decisions Are Not the Best Advice: Making Adherence-Aware Recommendations*
with J. Grand-Clément
Management Science, 2024

7. *Hospital-wide Inpatient Flow Optimization*
with D. Bertsimas
Management Science, 2023
    - 2022 POMS College of Healthcare Operations Management Best Paper Award, Winner
    - 2021 Canadian Healthcare Optimization Workshop Best Paper Competition, Winner
    - 2020 INFORMS Doing Good with Good OR Student Competition, Honorable Mention

6. *Robust Combination Testing: Methods and Application to COVID-19 Detection*
with S. Jain, J. Jónasson, and K. Ramdas
Management Science, 2023

5. *Predicting Inpatient Flow at a Major Hospital using Interpretable Analytics*
with D. Bertsimas and Beth Isreal Deaconess Medical Center (J. Stevens, M. Tandon)
Manufacturing & Service Operations Management, 2021

4. *From Predictions to Prescriptions: A Data-Driven Response to COVID-19*
with D. Bertsimas et al.
Health Care Management Science, 2021
   - 2020 INFORMS Pierskalla Award, Winner

Analytics for sustainable operations

3. *Optimizing the Path Towards Plastic-Free Oceans*
with D. den Hertog, B. Song, and The Ocean Cleanup (Yannick Pham, Bruno Sainte-Rose)
Operations Research, 2024
   - 2026 ENRE Best Publication Award in Environment and Sustainability
   - 2025 MSOM Responsible Research Award, Runner-up
   - 2025 Best OM Paper in Operations Research, Finalist
   - 2025 INFORMS Innovative Applications in Analytics Award, Third Place
   - (B. Song) 2025 M&SOM Best Student Paper Award, Winner
   - (B. Song) 2024 INFORMS Doing Good with Good OR Student Competition, Honorable Mention
   - (B. Song) 2024 POMS College of Sustainable Operations Student Paper Competition, Winner
   - (B. Song) 2023 INFORMS Transportation Society and Logistics Best Student Paper Competition, Honorable Mention

2. *Direct Optimization across Computer Generated Reaction Networks Balances Materials Use and Feasibility of Synthesis Plans for Molecule Mibraries*
with H. Gao, T. Struble, C. Coley, and K. F. Jensen
Journal of Chemical Information and Modeling, 2021

1. *A Tractable Numerical Strategy for Robust MILP and Application to Energy Management*
with L. El Ghaoui, D. Faille and D. Kiner
IEEE 55th Conference on Decision and Control, 2016

Honors and Awards

Research awards

- **ENRE Best Publication Award in Environment and Sustainability**, Winner, *2026*
- **MSOM Responsible Research Award**, Runner-Up, *2025*
- **Best OM Paper in Operations Research**, Finalist, *2025*
- **INFORMS Innovative Applications in Analytics Award**, Third Prize, *2025*
- **INFORMS Data Mining and Decision Analytics Theoretical Paper Competition**, Winner, *2024*
- **LBS Junior Faculty Research Award**, *2023*
- **POMS College of Healthcare Operations Management Best Paper Award**, Winner, *2022*
- **INFORMS Data Mining Section Best Paper**, Finalist, *2021.*
- **Canadian Healthcare Optimization Workshop Best Paper Competition**, Winner, *2021*
- **INFORMS M&SOM Data-Driven Challenge**, Winner, *2020*
- **INFORMS Pierskalla Award**, Winner, *2020*
- **INFORMS George E. Nicholson Student Paper Competition**, Winner, *2020*
- **INFORMS George B. Dantzig Dissertation Award**, Honorable Mention, *2020*

- **INFORMS Doing Good with Good OR Student Competition**, Honorable Mention, *2020*
- **INFORMS Computing Society Student Paper Award**, Winner, *2019*

Research awards by student co-authors

- Yupeng Wu: **INFORMS Computing Society Student Prize**, Honorable Mention, *2025*
- Baizhi Song: **M&SOM Best Student Paper Award**, Winner, *2025*
- Baizhi Song: **INFORMS Doing Good with Good OR Student Competition**, Honorable Mention, *2024*
- Baizhi Song: **POMS College of Sustainable Operations Student Paper Competition**, Winner, *2024*
- Baizhi Song: **INFORMS Transportation Science and Logistics Best Student Paper Competition**, Honorable Mention, *2023*
- Liangyuan Na and Kimberley Villalobos Carballo: **INFORMS Doing Good with Good OR Student Competition**, Finalist, *2023*

Teaching & Service awards

- **Poets & Quants 40 Under 40 Best MBA Professors**, *2025*
- **Operations Research Meritorious Service Award**, *2024*
- **INFORMS Journal on Computing Meritorious Reviewer Award**, *2024*

Student Advising

Doctoral students

1. Baizhi Song, London Business School, 2021–2026 (first placement: Assistant Professor, Wisconsin School of Business, University of Wisconsin-Madison)
2. Yupeng Wu, London Business School, 2023–
3. Erfan Amani Bani, London Business School, 2025–

Master students

Mohammed Yahya Soali (2021 → Columbia IEOR Ph.D.), Killian Dengreville (2022 → Corps des Mines), Christian Kotait (2022), Arthur Divanovic (2023), Pierre Aguié (2024).

Teaching Experience

TC03: Business Analytics (MBA level) Summer 2021, 2022, 2023, 2024, 2025
Lecturer

CM16: Business Analytics (MSc level) Fall 2024
Lecturer

P194: MSO PhD Seminar Advanced Optimization (PhD level) Summer 2023, 2025
Lecturer Topic: Duality theory, Optimization algorithms, Discrete optimization

P241: Emerging Topics in MSO (PhD level) Fall 2022
Lecturer Topic: Optimization under data uncertainty

Kaufman Teaching Certificate Summer 2019
Participant

15.094: Robust Modeling, Optimization and Computation (MSc/PhD level) Spring 2019
Teaching Assistant Instructors: Dimitris Bertsimas, Dick den Hertog

15.S41: Software Tools for Business Analytics (BSc level) Winter 2019
15.S60: Computing in Optimization and Statistics (MSc/PhD level)
Instructor

15.060: Data, Models, Decision (MBA level) Fall 2017
Teaching Assistant. Instructor: Juan-Pablo Vielma

15.572: Analytics Lab (MBA/MBAn level) Fall 2017
Team mentor

Invited Talks

Research seminars: MIT Sloan School of Management – Operations Management (2026), Stanford Graduate School of Business – Operations, Information & Technology (2026), Imperial College – Department of Mathematics (2025), EPFL – Management of Technology & Entrepreneurship Institute (2025), Aarhus University – Department of Economics and Business Economics (2024), MIT – Operations Research Center (2024), University of Maryland Robert H. Smith School of Business – Decision, Operations & Information Technologies (2024), Yale School of Management – Operations (2024), University of Wisconsin-Madison – Department of Industrial & Systems Engineering (2024), UCLA Anderson School of Management – Decisions, Operations, and Technology Management (2023), HEC – Information Systems and Operations Management (2023), Bayes Business School – Operations Management (2023), Imperial College Business School – Operations & Analytics (2022), Purdue University Krannert School of Management – Supply Chain and Operations Management (2022), GeorgiaTech – Industrial and Systems Engineering (2022), University of Toronto Rotman – Young Scholar Seminar (2022), Bilkent University – Department of Industrial Engineering (2021), European Technology & Operations Management Webinar (2021), Carnegie Mellon University Tepper School of Business – Operations Research (2020), INSEAD – Decision Sciences (2020), HEC Montréal – Decision Science (2020), University of Toronto Rotman – Operations Management and Statistics (2020), Columbia University – Operations Research and Industrial Engineering (2020), Cornell University – Information Engineering and Operations Research (2020), USC Marshall School of Business – Data Science and Operations (2020), IESE – Production, Technology and Operations Management (2019), Tilburg University School of Economics and Management – Econometrics and Operations Research (2019), London Business School – Management Science and Operations (2019), McGill Desautels Faculty of Management – Operations Management (2019), EPFL – Management of Technology & Entrepreneurship Institute (2019).

Single-track workshops: Workshop in Innovation and Learning (HEC: 2026, LBS: 2025, INSEAD: 2023), TTIC Workshop – "Data Driven Decision Processes: From Theory to Practice" (2024), European Technology and Operations Management Day – University of Cambridge (2024), Summer Workshop on Innovations in Management Science (2023), IMSI Workshop – "Predictive Analytics, Business Modeling and Optimization in Healthcare OM" (2023), TUM Heilbronn Workshop – "Data-Driven Resource Management" (2023), Dagstuhl Seminar – "Optimization at the Second Level" (2022).

Conferences: INFORMS Annual Meeting (2026, 2024, 2023, 2022, 2021, 2020, 2019, 2018, 2017), Conference on Integer Programming and Combinatorial Optimization (IPCO 2026), Joint Conference on Computational and Financial Econometrics and Computational and Methodological Statistics (CFE-CMS 2025), International Conference on Stochastic Programming (2025), MSOM Annual Conference (2022, 2021), POMS Annual Conference (2022, 2019), INFORMS Healthcare Conference (2021, 2019), Canadian Healthcare Optimization Workshop (2021), SIAM Conference on Optimization (2021), International Symposium on Mathematical Programming (2018).

Academic and University Service

Academic

- Co-initiator and co-organizer of London Operations Research Day (2024 at LBS, 95 registered participants; 2025 at Oxford, 120 registered participants; 2026 at Imperial).

- Area Chair for the 2025 MSOM Conference (Sustainability track, 160+ submissions) and for the 2025 annual conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO)
- Journal reviewer for *Management Science*, *Operations Research*, *Manufacturing & Service Operations Management*, *Mathematical Programming*, *Mathematics of Operations Research*, *INFORMS Journal on Computing*, *SIAM Journal on Optimization*, *INFORMS Journal on Optimization*, *European Journal of Operational Research*, *Production and Operations Management*, *Journal of the American Statistical Association*, *Communications Medicine*, *Transactions in Operations Research*.
- Conference reviewer for M&SOM Healthcare SIG Conference (2026, 2024, 2023, 2022), M&SOM Service Management SIG Conference (2022), Conference on Integer Programming and Combinatorial Optimization (2023), IEEE Conference on Decision and Control (2023).
- Grant reviewer for the European Research Council (Postdoctoral Fellowships: 2024), the Chilean National Agency for Research and Development, the Israel Science Foundation (Personal Research Grants: 2025).
- Committee member for the MSOM Responsible Research Best Paper Award (2026).
- Judge for the INFORMS G. Nicholson Best Student Paper Award (2026, 2025), INFORMS Public Sector OR Best Paper Award (2025), M&SOM Data-Driven Challenge (2025), M&SOM Student Paper Competition (2025, 2024), INFORMS Data Mining Best Paper Award (2023).

London Business School

- PhD coordinator (2025–)
- Co-organizer of the MSO seminar series (2022–)
- Grader for Ph.D. 2nd-year paper: Alireza AmaniHamedani (2022), Yuhang Du (2023), Wani Zhang (2024), Federico Tiberga (2026)
- Committee member for Ph.D. transfer examination: Alireza AmaniHamedani (2023), Yuhang Du (2024)
- Confidential Advisor (2023–)

Industry Experience

Freelance Consultant — London, UK
Consultant and instructor in Analytics — 2018–

Predilex — Paris, France
Scientific Advisor — Sep 2018–Mar 2020

Ventech — Paris, France
Venture Capital Analyst — Oct 2015–Sep 2016

University of California, Berkeley — Berkeley, CA
Électricité de France — Châtoux, France
Visiting Student Researcher, advised by Laurent El Ghaoui — Mar 2015–Aug 2015

VINCI Energies group — Weil-am-Rhein, Germany
Assistant Project Manager — Jul 2014–Aug 2014

Skills

Languages	French (native), English (fluent), German (advanced)
Programming	Julia, Python, R, Matlab, Caml, SQL, MongoDB